

TIRTH GOTI

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EDUCATION

GLS University — Faculty of Engineering & Technology

2023 – 2027

B.Tech, Computer Science & Engineering (AI/ML) | CGPA: 8.5/10

Sanskartirth Gyanpeeth, Surat

Class XII (GSEB) | 76.76%

Sanskartirth Gyanpeeth, Surat

Class X (GSEB) | 93%

TECHNICAL SKILLS

Programming Languages: Python, SQL

Machine Learning: Scikit-learn, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib

Computer Vision: YOLOv11, Image Segmentation, Image Processing

Generative AI: LangChain, Gemini API, Prompt Engineering

Databases: MySQL, SQLite, Supabase

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook

PROJECTS

FixMyRoad — AI-Powered Road Damage Detection & Repair Cost Estimation

Python • YOLOv11 • FastAPI • OpenCV • Supabase • SQLite

GitHub: github.com/tirthgoti/FixMyRoad

- Developed an AI-powered road inspection system that detects potholes and road damage from uploaded images using computer vision.
- Curated and annotated a custom dataset of 2,000 road images using Roboflow, then trained a YOLOv11 segmentation model achieving 91.7% pothole detection accuracy.
- Designed RESTful APIs using FastAPI for image upload, damage detection, and report generation.
- Built a backend using Supabase and SQLite to manage user authentication, road damage reports, and complaint history.
- Researching monocular depth estimation to estimate pothole dimensions for repair cost and material estimation (planned extension).
- Designed an authority dashboard for monitoring reported road damages and prioritizing repairs by severity and location.

AI Resume Analyzer with ATS Score & Job Matching

Python • NLP • Scikit-learn • FastAPI • SQLite

GitHub: [github.com/tirthgoti/AI Resume Analyzer](https://github.com/tirthgoti/AI_Resume_Analyzer)

- Developed an AI-powered resume analysis system that extracts candidate skills, education, and experience from PDF resumes.
- Implemented ATS-style resume scoring and job description matching using natural language processing techniques.
- Built keyword extraction and skill gap analysis modules to recommend improvements based on job requirements.
- Designed backend APIs for resume parsing, candidate profile management, and recommendation generation.

Intelligent PDF Assistant using Retrieval-Augmented Generation (RAG)

Python • LangChain • Gemini API • FastAPI • Streamlit

GitHub: [github.com/tirthgoti/PDF Assistant](https://github.com/tirthgoti/PDF_Assistant)

- Developed a document assistant that answers natural language questions from uploaded PDF documents.
- Implemented Retrieval-Augmented Generation using LangChain and FAISS to retrieve relevant context before generating responses.
- Integrated Gemini API to generate context-aware answers based on retrieved document chunks.
- Designed a document retrieval pipeline using vector embeddings for efficient information retrieval.

RELEVANT COURSEWORK

Machine Learning, Artificial Intelligence, Deep Learning, Database Management Systems

ACHIEVEMENTS

- Awarded 100% Tuition Fee Waiver at GLS University based on academic excellence.
- Received 100% Scholarship in Classes XI and XII for outstanding academic performance.
- Participated in college technical events, AI/ML competitions, and hackathons.

CERTIFICATIONS

- Foundation of Artificial Intelligence — Green Skill Initiative